

Problem 1: Put a penny on each gray square of strip A. A jump: a peg hops over a peg right next to it, in a straight line, into the empty square just beyond, and the peg it hopped over comes off the strip. Make one jump. Draw an X in the square where the peg lands.

Strip A

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Problem 2: On strips B, C and D, put pennies on the gray squares and try to make a jump. Circle *possible* or *not possible* for each strip. On strip D, draw an X in every square a peg could land on with one jump.

Strip B

circle one: possible not possible

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Strip C

circle one: possible not possible

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Strip D

circle one: possible not possible

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Problem 3: Work in pairs: one of you jumps, the other records, and you swap after each strip. On each of strips E to H, put pennies on the gray squares. Jump until no jump is possible, in either direction. Write the number of pegs left. Then set the strip up again and jump in a different order. Write the fewest pegs you managed to leave.

Strip E

pegs left, first try: _____ fewest: _____

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Strip F

pegs left, first try: _____ fewest: _____

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Problem 3, continued: strips G and H.

Strip G

pegs left, first try: _____ fewest: _____



Strip H

pegs left, first try: _____ fewest: _____



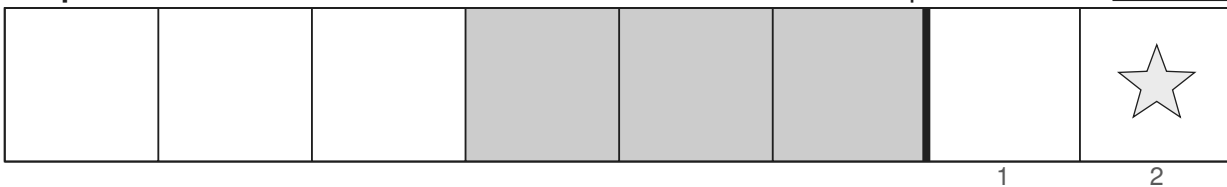
Problem 4: On strip I the thick line is the river, and the squares to the left of it are the home side. Put pennies on the two gray squares. Cross the river with one jump. Draw an X where the peg lands. Strips J, K and L start with three, four and six pegs on the home side. Keep working in pairs: one jumps, the other watches for the farthest square. For each strip, put pennies on the gray squares, jump as many times as you like, and try to get a peg onto square 2, the square with the star. Write the farthest square any peg reached, 1 or 2.

Strip I



Strip J

farthest square reached: _____



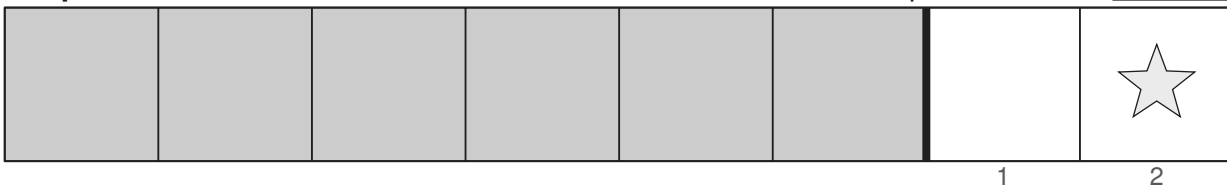
Strip K

farthest square reached: _____



Strip L

farthest square reached: _____



Problem 5: On strip M, draw a dot in up to six home squares to make a starting position for the person next to you. Swap pages. Put pennies on the dots you were given, play, and write the farthest square any peg reached.

Strip M

farthest square reached: _____

							★
						1	2

Problem 6: Strip N has a value printed in the corner of each square. The total of a position is the sum of the values of the squares that have pegs on them. For each row of the table: put pennies on the squares listed, write the total, make the jump described, and write the new total. In the last column write *up*, *same* or *down*.

Strip N

1	1	2	3	5	8	13	21 ★
						1	2

pegs on the squares valued ...	total before	total after	up, same or down
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all six home squares; make the only jump that is possible	_____	_____	_____
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2 and 3; jump to the left	_____	_____	_____
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2 and 3; jump to the right	_____	_____	_____
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8 and 13, one on each side of the river; jump to the left	_____	_____	_____
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Problem 7: Use strip N again. Try to find a jump that makes the total go up: start from any two pegs you like, and try jumps to the right and jumps to the left. Write three of your tries in the table. Then circle your answer: did any jump make the total go up? yes no

from the square valued ... over ... to ...	total before	total after	up, same or down
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_____ over _____ to _____	_____	_____	_____
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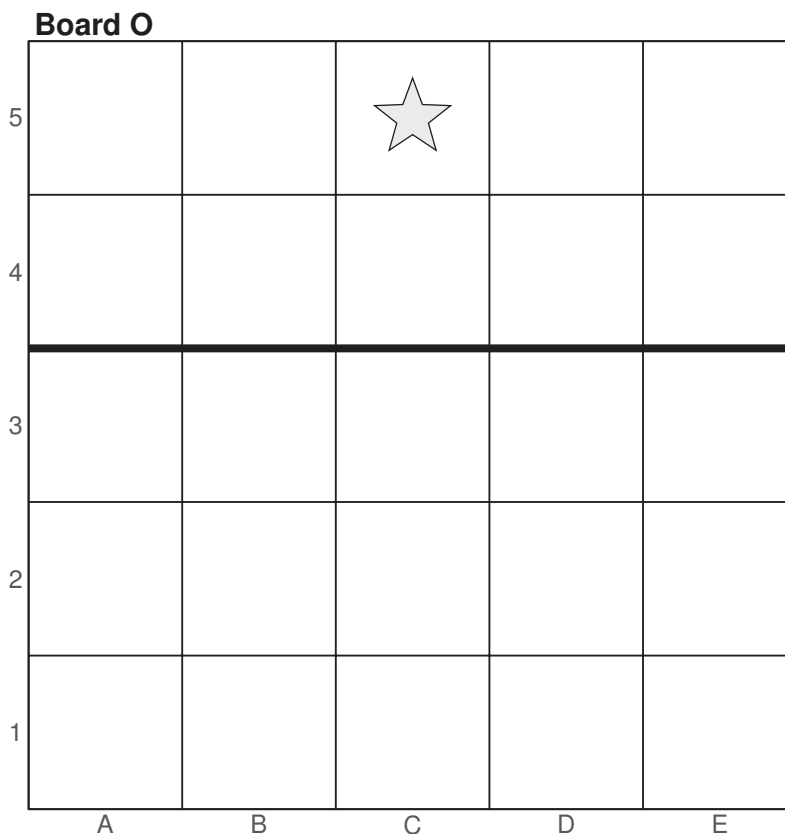
_____ over _____ to _____	_____	_____	_____
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_____ over _____ to _____	_____	_____	_____
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Problem 8: When a peg jumps to the right on strip N, it leaves two squares and lands on the next one. Write the three values of one jump to the right as a sum: $\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$. Tell the adult at your table why the total is exactly the same after every jump to the right, and why it goes down after every jump to the left.

Problem 9: Write the total of the six home squares of strip N: $\underline{\hspace{1cm}}$ and the value of square 2: $\underline{\hspace{1cm}}$. Tell the adult why a peg can never land on square 2 of strip N, whatever jumps you make, even if you start with fewer pegs. Then write your reason in one or two sentences.

Problem 10: Board O is a grid with the river across it. Squares are named by column letter and row number, like C3. On a board, jumps go up, down, left or right, never diagonally. Put pennies on C3 and C2. Cross the river with one jump. Write the name of the square where the peg lands: $\underline{\hspace{1cm}}$



Problem 11: Each small picture shows a shape of four pegs on the home side of board O. Share the six shapes out around the table, one or two each. For each of your shapes, copy it onto board O with pennies and try to get a peg to the star at C5, jumping as many times as you like. Under the picture write *yes* if a peg reached the star and *no* if it never did. For a shape that works, also write its jumps in order, like C1 over C2 to C3. Before the table writes *yes* in the table results, a different person must replay those jumps on board O.

Shape 1

works: _____
jumps: _____

Shape 2

works: _____
jumps: _____

Shape 3

works: _____
jumps: _____

Shape 4

works: _____
jumps: _____

Shape 5

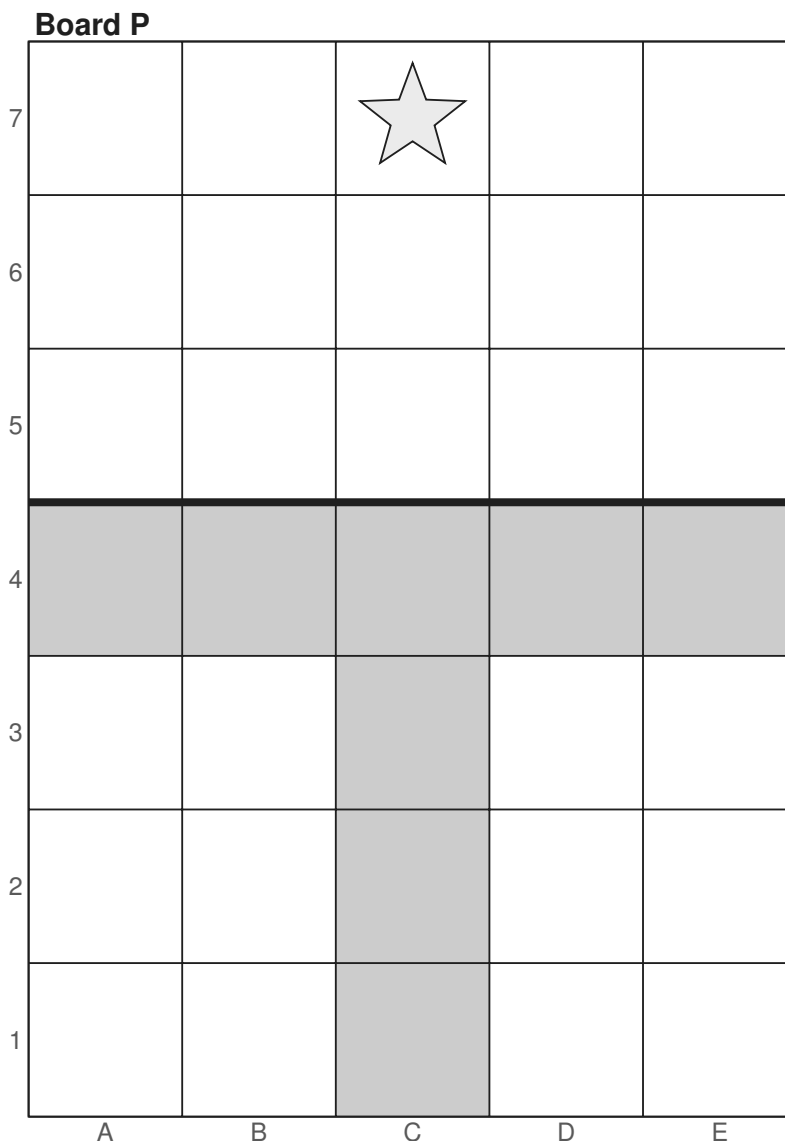
works: _____
jumps: _____

Shape 6

works: _____
jumps: _____

table results	Shape 1	Shape 2	Shape 3	Shape 4	Shape 5	Shape 6
reached the star (yes or no)						

Problem 12: Board P has eight pegs: five along the river and three below the middle one. The whole table works on one board. Put pennies on the gray squares. Find seven jumps that put a peg on the star at C7. Agree on each jump before anyone makes it, and take turns writing the jumps down: the square the peg jumps from, the square it jumps over, and the square it lands on. If you get stuck, set the pegs up again and write a fresh list.



Jump 1: from _____ over _____ to _____

Jump 5: from _____ over _____ to _____

Jump 2: from _____ over _____ to _____

Jump 6: from _____ over _____ to _____

Jump 3: from _____ over _____ to _____

Jump 7: from _____ over _____ to _____

Jump 4: from _____ over _____ to _____

Problem 13 (Extra): Strip Q is empty. Draw dots in five of its squares to make a pattern of five pegs. Pass the page to your neighbour, who must clear your pattern down to one peg with pennies, or try every order and show that it cannot be done. Your neighbour writes the fewest pegs left. Then, on strip R, draw a five-peg pattern that always gets stuck with three or more pegs left, and check it with pennies yourself.

Strip Q

fewest pegs left: _____

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Strip R

fewest pegs left: _____

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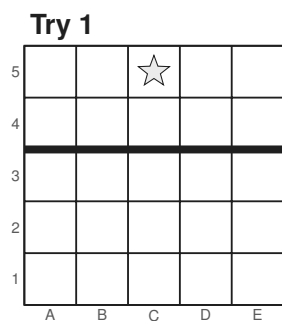
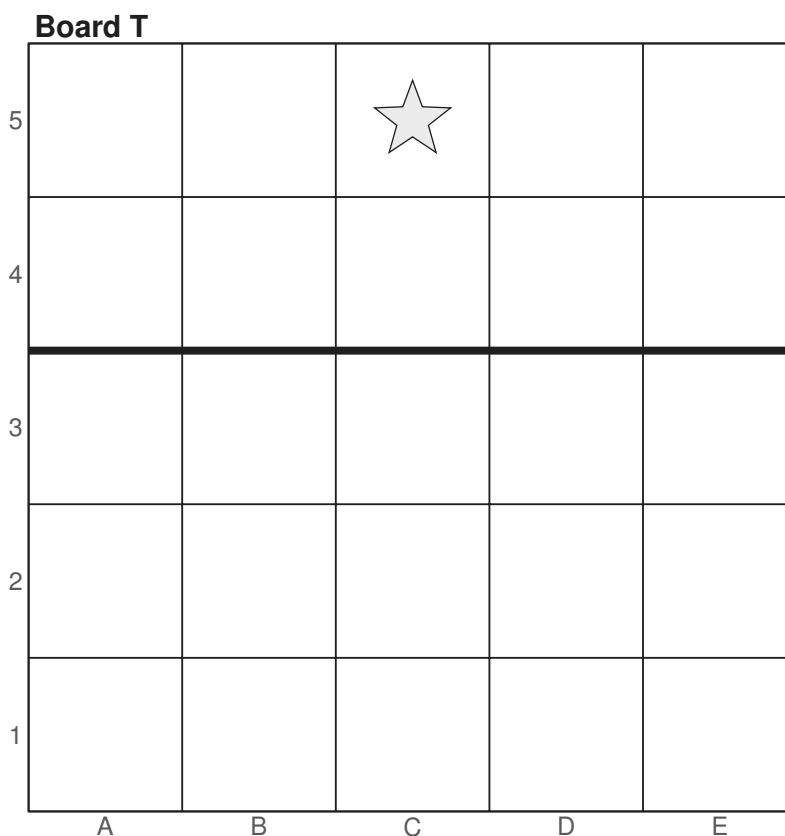
Problem 14 (Extra): Strip S has seven home squares and no values. Write values in its corners so that the pattern of strip N continues from the left: each value is the sum of the two values before it. Write the total of the seven home squares: _____ and the value of square 2: _____. Say why no peg can reach square 2 of strip S either.

Strip S

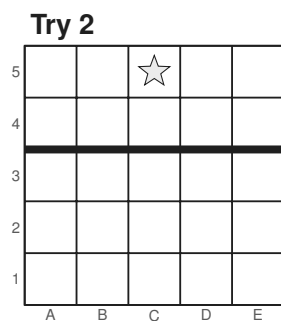
1

2

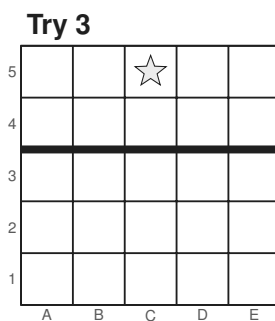
Problem 15 (Extra): Board T is empty. Put only three pennies on home squares of your choice and try to get a peg to the star at C5. Try at least four different placements. Record each one on a small board below: draw a dot in each square you used and write the highest row number any peg reached.



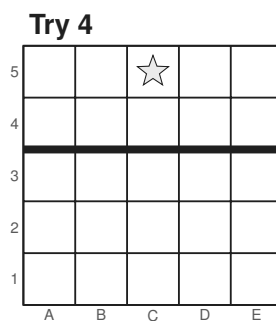
highest row: _____



highest row: _____

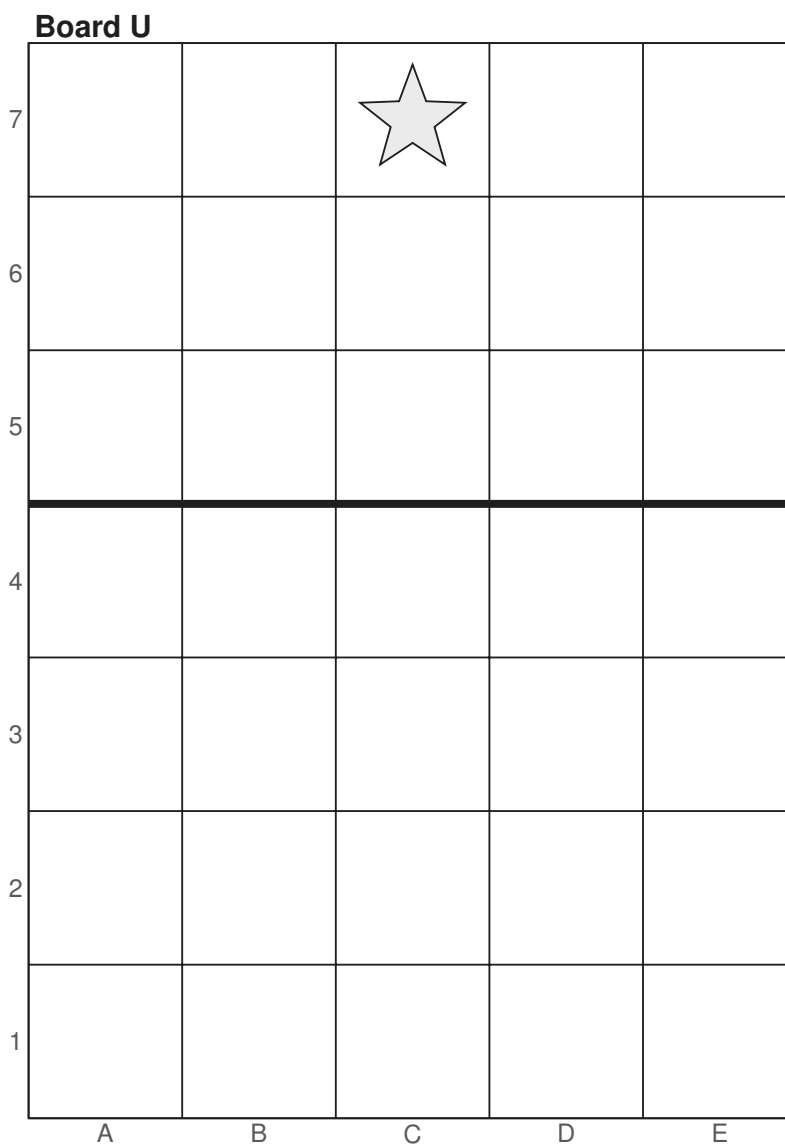


highest row: _____



highest row: _____

Problem 16 (Extra): Board U is empty. Find a set of eight home squares, different from the shape of board P, from which a peg can reach the star at C7. Test it with pennies. Draw a dot in each of the eight squares of a set that works, and list its seven jumps.



Jump 1: from _____ over _____ to _____

Jump 5: from _____ over _____ to _____

Jump 2: from _____ over _____ to _____

Jump 6: from _____ over _____ to _____

Jump 3: from _____ over _____ to _____

Jump 7: from _____ over _____ to _____

Jump 4: from _____ over _____ to _____